

PRST Class (29/09/2020)

Moment generating function (MGF)

$$M(t) = E(e^{tX})$$

X is discrete $M(t) = \sum_{x_i \in S} e^{tx_i} p(x_i)$

X is cont $M(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$

$$e^{tX} = \sum_{r=0}^{\infty} \frac{(Xt)^r}{r!} = \sum_{r=0}^{\infty} \frac{t^r X^r}{r!}$$

$$E(e^{tX}) = \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r' \quad \left| E(X^r) = \mu_r' \right.$$

$$M(t) = E(e^{tX})$$

$$\frac{dM(t)}{dt} = E(e^{tX} X)$$

$$\left. \frac{dM(t)}{dt} \right|_{t=0} = E(X) = \mu$$

$$\left. \frac{d^2 M(t)}{dt^2} \right|_{t=0} = E(X^2 e^{tX}) \Big|_{t=0} = E(X^2) = \mu_2'$$

$$\left. \frac{d^r M(t)}{dt^r} \right|_{t=0} = E(X^r) = \mu_r'$$

$$\left. \frac{d}{dt} \right|_{t=0}$$

Ex 1

$$M(t) = (q + pe^t)^n$$

$$| q = 1-p$$

$$E(X) = ?, \quad E(X^2) = ?$$

$$= \mu \quad \mu_2'$$

$$\mu = \left. \frac{dM(t)}{dt} \right|_{t=0} = n (q + pe^t)^{n-1} \cdot p \Big|_{t=0} = np$$

$$\mu_2' = \left. \frac{d^2 M(t)}{dt^2} \right|_{t=0} = np^2 (n-1) (q + pe^t)^{n-2} \Big|_{t=0}$$

$$= n(n-1)p^2$$

Result

$$E(e^{tY})$$

$$= E(e^{t(ax+b)})$$

$$= E(e^{tb + taX})$$

$$= E(e^{tb} \cdot e^{atX})$$

$$= e^{tb} M(at)$$

$$| \begin{array}{l} X, Y = aX + b \\ \text{mgf} \rightarrow M(t) \end{array}$$

Joint distribution of two variables.

four unbiased coins are tossed.

$$\omega_1 = HHHH$$

$$\omega_2 = HHHT$$

X : number of H's in an outcome

Y : length of the run

$\omega_1 =$
 $\omega_2 = \text{HHHT}$
 $\omega_3 = \text{HHTH}$
 $\omega_4 = \text{HHTT}$

Y : length of the longest run of H's

$\omega_{15} = \text{TTTH}$
 $\omega_{16} = \text{TTTT}$

$X \backslash Y$	0	1	2	3	4	
0	$\frac{1}{16}$	0	0	0	0	$\frac{1}{16}$
1	0	$\frac{4}{16}$	0	0	0	$\frac{4}{16}$
2	0	$\frac{3}{16}$	$\frac{3}{16}$	0	0	$\frac{6}{16}$
3	0	0	$\frac{2}{16}$	$\frac{2}{16}$	0	$\frac{4}{16}$
4	0	0	0	0	$\frac{1}{16}$	$\frac{1}{16}$
	$\frac{1}{16}$	$\frac{7}{16}$	$\frac{5}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	1

$$P(X=0, Y=0)$$

$$P(X=0, Y=1)$$

$$P(X=1, Y=0)$$

$$P(X=1, Y=1) =$$

$$(X=x \mid P(X=x))$$

$$X=x, Y=y \mid P(X=x, Y=y)$$

$$P(X=4)$$

$$P(Y=0)$$

$$X = \{0, 1, 2, 3, 4\}$$

$X=x$	$P(X=x)$
0	$P(X=0) = \frac{1}{16}$
1	$P(X=1) = \frac{4}{16}$
2	$P(X=2) = \frac{6}{16}$
3	$P(X=3) = \frac{4}{16}$
4	$P(X=4) = \frac{1}{16}$

Marginal dist of X .

$$P(X=0)$$

$$\begin{aligned}
 &= P(X=0, Y=0) \\
 &+ P(X=0, Y=1) \\
 &+ P(X=0, Y=2) \\
 &+ P(X=0, Y=3) \\
 &+ P(X=0, Y=4)
 \end{aligned}$$

$$Y \mid P(Y=y)$$

$$1 \text{ or } 0$$

Y	$P(Y=Y)$	
0	$P(Y=0) = \frac{1}{16}$	$P(Y=0)$
1	$P(Y=1) = \frac{7}{16}$	$= P(Y=0, X=0)$
2	$P(Y=2) = \frac{5}{16}$	$+ P(Y=0, X=1)$
3	$P(Y=3) = \frac{2}{16}$	$+ P(Y=0, X=2)$
4	$P(Y=4) = \frac{1}{16}$	$+ P(Y=0, X=3)$
		$+ P(Y=0, X=4)$

/
marginal dist of Y .